

Nikolaos Xiros

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About

As a recent graduate in Electrical and Computer Engineering at the National Technical University of Athens, I'm passionate about Artificial Intelligence and Machine Learning, with a particular focus on Multimodal Learning and NLP. I enjoy studying novel architectures, optimization methods, and the behavior of existing models. Beyond the technical aspects, I'm deeply interested in understanding how AI systems interact with society by exploring their fairness, transparency, and ethical implications. Through hands-on research projects, I continuously develop my technical skills and deepen my understanding of AI's theoretical foundations to address these complex challenges.

Education

- National Technical University of Athens** Oct 2020 – Oct 2025
Integrated Master's (MEng) in Electrical and Computer Engineering Athens, Greece
 - GPA: 9.08/10 (High Distinction – Ranked top 7% of class)
 - Thesis: *Refining CLIP Visual and Language Representations via Self-Supervised Visual Structure*
 - Supervisor: **Prof. Alexandros Potamianos**
- Technical University of Denmark (DTU)** Aug 2023 – Jan 2024
Erasmus+ Exchange Program Lyngby, Denmark

Research Experience

- Institute for Language and Speech Processing (ILSP)** Oct 2025 – Present
Associate Researcher (previously Research Intern) Athens, Greece
 - Researching novel router architectures for Mixture of Experts models.
 - Researching interpretability and explainability techniques in multimodal settings.
 - Training, evaluating and deploying large-scale ML models on HPC infrastructure.
- University of Science and Technology of China (USTC)** Jul 2025 – Aug 2025
Future Scientist Exchange Program (FuSEP) – Summer Research Hefei, China
 - Participated in a 5-week international research program at USTC, collaborating with leading researchers.
 - Worked on data-driven and neural network-based modeling to capture and control the dynamics of a novel soft robotic system.

Skills

- Programming:** Python, C/C++, SQL, Web Development, Bash, Git, Operating Systems
- Machine Learning:** PyTorch, HuggingFace, NumPy, OpenCV, Megatron-LM, Deepspeed

Projects

- IEEE NTUA Student Branch** Sep 2022 – Sep 2024
Member Athens, Greece
 - Active volunteer and project contributor in the IEEE student branch at NTUA.
 - Participated in the Bosch Future Mobility Challenge, developing robotics and AI modules for autonomous driving tasks.

Honours and Awards

- Mathematics Competitions – Hellenic Mathematical Society (Thales & Euclid distinctions)
- Physics Competition – Union of Greek Physicists (6th Nationwide, 30th National Competition)
- Graduation with High Distinction (GPA 9.08/10) – NTUA ECE

Languages

Greek: Native

English: C2 (Proficient) – Listening, Reading, Speaking, Writing

- [S.1] Alexis Filippakopoulos, **Nikolaos Xiros**, Elias Kallioras, Efthymios Georgiou, Alexandros Potamianos. (2025). **Segregate, Refine, Integrate: Decomposing Multimodal Fusion for Sentiment Analysis**. Manuscript submitted for publication in *Interspeech 2026*.